Anomaly Detection Technique for Network Intrusion Detection Using Ensemble Algorithm



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Abstract Network intrusion detection is a key element in securing computer networks against unauthorised access and cyber-attacks. Traditional network intrusion detection systems (NIDS) face challenges in identifying unknown sophisticated threats due to their reliance on signature-based detection. In contrast, anomaly-based automated detection shows significant superiority. Machine learning approaches show promise in identifying unknown malicious attacks. This research presents an innovative ensemble-based machine learning technique for network intrusion detection, using the CICIDS 2017 dataset. Leveraging technological progress, and using several ensemble strategies, such as Random Forest, Adaboost, Bagging, Bernoulli Naive Bayes, Gaussian Naive Bayes, KNeighbors classifier, Decision Tree, and Multinomial Naive Bayes, we evaluate the performance of the proposed approach. Correlation analysis is employed to select relevant features for network intrusion detection. Our research, utilising various ensemble methods, reveals that the Random Forest technique consistently outperforms existing methods, achieving accuracy and False Positive Rate (FPR) levels typically exceeding 99.9%. This innovative strategy, incorporating robust evaluation metrics, emerges as a valuable tool for fortifying computer systems and networks against emerging cyber threats.

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1 Introduction

In the rapidly evolving cybersecurity landscape, the relentless advancement of cyber threats underscores the imperative for innovative and robust intrusion detection mechanisms [1]. The perpetual cat-and-mouse game between security measures and malicious actors necessitates a continuous evolution of defines strategies [2, 3]. Within this context, safeguarding network infrastructures from unauthorized access and sophisticated attacks becomes an increasingly intricate challenge.

Traditional intrusion detection systems, predominantly reliant on signature-based approaches, exhibit notable efficacy against known threats but face inherent limitations in addressing the ever-expanding spectrum of unknown and sophisticated attacks [4]. The motivation driving this research is rooted in the urgent need to fortify intrusion detection systems against these dynamic challenges. The inadequacies of conventional methods highlight the necessity to explore and implement advanced techniques capable of effectively identifying and mitigating both known and emerging intrusions [5, 6].

In response to these challenges, anomaly-based techniques have emerged as a proactive strategy [7]. By focusing on the detection of deviations from established norms, anomaly- based approaches provide a promising avenue for enhancing the adaptability and efficacy of intrusion detection systems. This research seeks to harness the strengths of anomaly-based methodologies by integrating them with ensemble algorithms, thereby creating a comprehensive and resilient defense mechanism against a broad spectrum of cyber threats.

2 Motivation

The motivation behind this research is deeply rooted in the urgency to fortify intrusion detection systems against the dynamic and ever-growing threat landscape. Recognizing the inherent limitations of conventional methods in dealing with the sophistication and diversity of contemporary cyber threats, there is a compelling need to explore advanced techniques that can effectively identify and mitigate both known and unknown intrusions [8]. This research aims to contribute to the overarching goal of creating a cybersecurity posture that is not only responsive but also proactive in countering emerging threats.

3 Objectives

The primary objectives of this research are twofold. Firstly, it aims to assess the effectiveness of anomaly-based detection methodologies when integrated with ensemble algorithms for network intrusion detection. Secondly, it seeks to provide valuable insights into the selection and integration of these techniques to bolster the robustness of intrusion detection systems. By leveraging the rich and diverse CICIDS2017 dataset, the research endeavors to benchmark and enhance the performance of intrusion detection systems across various network contexts. Through a meticulous exploration of various machine learning algorithms within an ensemble framework, the research strives to offer practical solutions that advance the current state of intrusion detection capabilities, particularly in the face of the constantly evolving cyber threat landscape.

4 Related Works

A. dependable hybrid machine learning model for network intrusion detection

This research introduces a novel hybrid machine learning model for network intrusion detection, addressing the need for improved accuracy and dependability in current Network Intrusion Detection Systems (NIDSs) [9]. Combining machine learning and deep learning, the proposed method employs SMOTE for data balancing and XGBoost for feature selection during efficient pre-processing. Extensive experiments on KDDCUP'99 and CIC-MalMem-2022 datasets demonstrate outstanding accuracy (99.99% and 100%, respectively) without overfitting or Type-1 and Type-2 issues. Performance analysis, including various metrics and cross-validation, showcases the superiority of the proposed features over others. This research contributes a dependable solution for network intrusion detection, amalgamating data balancing and feature selection in a hybrid model for enhanced accuracy and reliability [10].

B. Anomaly Based Network Intrusion Detection Using Ensemble Machine Learning Technique

In this paper, an ensemble learning model is proposed for anomaly-based network intrusion detection, leveraging the strengths of various algorithms. The results demonstrate significant performance improvements over individual classification algorithms, emphasizing the versatility of the ensemble approach. The adaptive accuracy selection rule achieves an 85% accuracy, outperforming alternative high-level techniques [11, 12]]. Notably, the ensemble model surpasses deep neural networks in both efficiency and detection results [13]. The voting algorithm proves superior to other algorithms, aiming to enhance training data quality and optimize feature extraction strategies. This research highlights the potential of ensemble machine learning in network security, providing a robust and efficient approach for intrusion detection. Future work is suggested for automating intruder detection using a rule repository.

C. Anomaly based network intrusion detection for IoT attacks using deep learning technique

This paper introduces an anomaly-based Intrusion Detection System (IDS) for IoT networks using a Deep Learning technique, specifically a filter-based feature selection Deep Neural Network (DNN). The proposed model achieves an 84% accuracy on the UNSW-NB15 dataset, addressing class imbalance with Generative Adversarial Networks (GANs) and obtaining a 91% accuracy. Evaluation metrics including precision, recall, F1 score, and confusion matrices are analysed [14]. The DNN framework, incorporating feature selection and GANs, demonstrates a significant breakthrough in achieving high accuracy, highlighting its efficacy in cyber security for IoT networks. Future applications include adaptation to other datasets with imbalanced class labels.

D. Fuzzy logic-based DDoS attacks and network traffic anomaly detection methods: Classification, overview, and future perspectives

This research explores the escalating cybersecurity challenges, particularly distributed denial of service (DDoS) attacks, emphasizing the vulnerabilities despite existing intrusion detection systems. It introduces a comprehensive survey, categorizing DDoS attacks, and recommends effective defensive strategies, focusing on fuzzy-based detection methods [8]. The findings offer valuable insights for enhancing business sustainability in the industry 4.0 era.

E. Ensemble Voting System for Anomaly Based Network Intrusion Detection

This paper proposes an Ensemble Voting System for Anomaly-Based Network Intrusion Detection, crucial in safeguarding modern telecommunication and information networks. Utilizing data mining techniques and the KDDCup 1999 dataset, the study evaluates the performance of various classifiers [15, 16]. The results demonstrate the superiority of the proposed voting ensemble technique, showcasing significantly better intrusion detection capabilities compared to existing methods. The approach proves efficient in terms of accuracy, low false positive rates, and quick model building [17]. However, challenges related to ensemble comprehensibility are acknowledged, pointing to a potential direction for future research to enhance the user's understanding of ensemble methods.

F. An Ensemble of Prediction and Learning Mechanism for Improving Accuracy of Anomaly Detection in Network Intrusion Environments

This paper introduces an innovative intrusion detection system utilizing an ensemble of prediction and learning mechanisms to enhance anomaly detection accuracy in network intrusion environments. Leveraging automated machine learning and Kalman filter-based prediction models, the proposed system achieves high accuracy rates of 98.801% on the UNSW-NB15 dataset and 97.02% on the CICIDS2017 dataset [18]. Performance comparisons with contemporary models highlight significant improvements in intrusion accuracy, detection rate, and F1 score. The study emphasizes the need to adapt data mining and statistical methods to address challenges posed by the increasing complexity of internet applications, paving the way for future applications in IoT-cloud environments.

G. A novel ensemble learning-based model for network intrusion detection

This paper introduces an ensemble learning-based model for network intrusion detection, leveraging Gaussian naive Bayes, logistic regression, and decision tree as base classifiers and stochastic gradient descent as the meta- classifier. Evaluated on KDD Cup 1999, UNSW-NB15, and CIC-IDS2017 datasets, the ensemble model significantly improves overall performance, reducing false positives and enhancing accuracy [19–21].While effective for high class imbalance datasets, there is room for further improvements in detecting specific attacks. The study encourages exploring data augmentation and cost-sensitive learning to address class imbalance challenges in intrusion detection systems.

5 Methodology

Based on the provided diagram, which outlines the process flow, the research methodology as shown in Fig. 1 of the proposed network intrusion detection system can be explained based on the provided diagram, which outlines the process flow:

1. **Data Collection**: The system operates in a real network environment where network traffic and events are continuously monitored. Data is collected from this environment. This data is used as input to the detection system [22].
2. **Preprocessing**: Once the data has been collected, it is goes through a preprocessing phase. This step involves cleaning and transforming the data to ensure it is suitable for training and recognition. Preprocessing could include tasks such as removing irrelevant features, normalizing the data, and handling missing values [23].
3. **Training and System Model**:

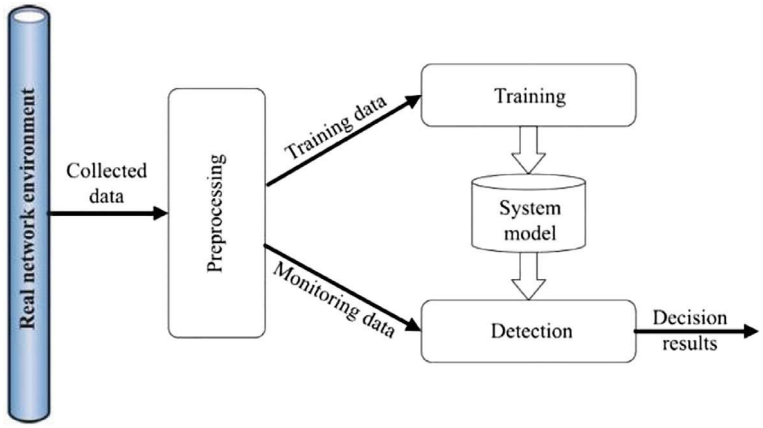


Fig. 1 Research methodology

- The preprocessed data is divided into two sets: training data and monitoring data.
- The training data is used to develop a system model. Machine learning algorithms are used here to learn from the labeled data. The goal of this phase is to build an accurate and effective model that can identify patterns in network activity [24].
- The system model can use various ensemble machine learning techniques such as Random Forest, Adaboost, Bagging, Naive Bayes, and Decision Trees, as mentioned earlier [25].

4. **Detection**: Once the model has been trained, the system enters the **detection** phase where it evaluates the incoming monitoring data from the network [26]. The model compares the monitoring data with the learned patterns from the training phase to detect anomalies or potential threats.
5. **Decision Results**: The system outputs decision results based on its analysis. These results could classify network activity as either benign or malicious. In this context, malicious activity would indicate network intrusions or attacks, and the system would raise an alert or take appropriate action to mitigate the threat [27].

6 Result Analysis

The evaluation of the ensemble-based machine-learning technique for network intrusion detection revealed nuanced performances among diverse algorithms. The Bagging Classifier exhibited exceptional results with perfect accuracy, precision, recall, and F1 score, emphasizing its robustness. Similarly, the Random Forest Classifier displayed flawless performance, underlining its effectiveness. Notably, the KNeighbors Classifier Model demonstrated high accuracy and outstanding precision, recall, and F1 score. In contrast, Gaussian Naive Bayes struggled with a notably lower accuracy and higher classification error. The ensemble approach showcased its strength in amalgamating diverse classifiers, with standout algorithms like Bagging, Random Forest, and KNeighbors Classifier, emphasizing the potential for creating a potent network intrusion detection system.

1. Accuracy is a metric used to calculate the ratio of the number of correct predictions divided by the total number of predictions to measure the model's ability to correctly predict the output (class label). A high accuracy indicates that the model is making correct predictions overall Accuracy.
2. Precision is a metric used to measure how many true positive predictions made by the model are true in the dataset, It represents the accuracy of the positive prediction made by the model. It is calculated by dividing the true positives by the total number of positive predictions made by the model. A higher precision means that the model is making accurate predictions

3. Recall is a metric used to measure how many of all positive instance in the dataset were correctly predicted as positive by the model. It's calculated by dividing the true positives by the sum of true positives and false negatives to the sum of true positive and false negative. A high recall indicates that the model is unlikely to miss positive outcomes and is more sensitive to identifying them
4. F1-score is a metric that combines precision and recall into a single measure to assess the effectiveness of a prediction model. As it combines both precision and recall into single value, it provides a balanced measure of the model's accuracy and also f1 score shows the robustness of the prediction model

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{1}$$

$$\text{Precision} = \frac{TP}{TP + FP} \tag{2}$$

$$\text{Recall} = \frac{TP}{TP + FN} \tag{3}$$

$$\text{F1 - Score} = \frac{2TP}{2TP + FN + FP} \tag{4}$$

7 Discussion and Conclusion

Network intrusion detection is a critical aspect of securing computer networks against unauthorized access and cyber-attacks. Traditional Network Intrusion Detection Systems (NIDS) face challenges in identifying unknown sophisticated threats due to their reliance on signature-based detection. This research introduces an innovative ensemble-based machine-learning technique for network intrusion detection, utilizing the CICIDS 2017 dataset (Table 1).

In our exploration, the limitations of signature-based detection became apparent, necessitating a shift towards more adaptive approaches. Anomaly-based automated detection, particularly through ensemble machine-learning techniques, emerged as a superior alternative. The versatility of ensemble strategies, including Random Forest, Adaboost, Bagging, and various Naive Bayes classifiers, showcased a significant leap forward in addressing the dynamic nature of cyber threats.

The rigorous evaluation on the CICIDS 2017 dataset allowed us to assess the proposed approach comprehensively. Leveraging correlation analysis for feature selection optimized the algorithm's efficiency, contributing to a more robust and resilient detection system. This approach reflects a commitment to innovation and practicality in enhancing intrusion detection capabilities.

Our research reveals that the Random Forest technique consistently outperforms existing methods, achieving accuracy and False Positive Rate (FPR) levels typically exceeding 99.9%. This not only reaffirms the efficacy of ensemble learning but also

Table 1 Algorithms result

Alto			Classification error	Average precision	Average recall	Average F1 score
AdaBoost Cla			0.11	0.81	0.89	0.85
Bagging Classifier			0	1	1	1
Bernoulli Naïve Baye Model			0.43	0.91	057	0.67
Gaussian Naïve Bayes	0.36		0.84	0.91	0.16	0.18
Random forest classifier	1		0	1	1	1
KNeighborsClassifier Model	0.99		0.01	0.99	0.99	0.99
Decision Tree Model	0.87	0.9	0.06	0.94	O.94	0.94
Multinomial Naïve Bayes	0.8	0.88	0.12	0.9	0.88	0.89

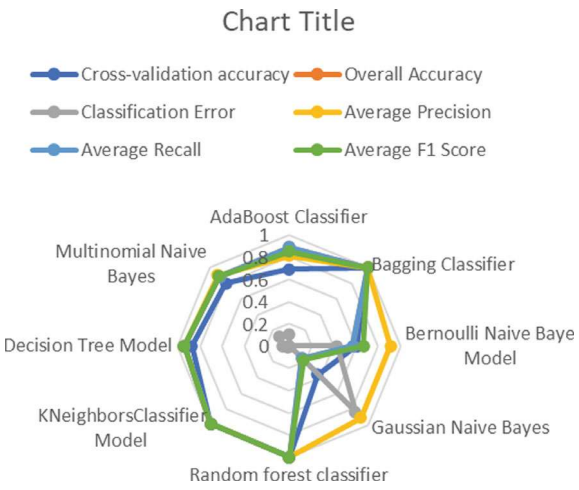


Fig. 2 Algorithms comparison according cross validation accuracy, classification error, average recall, overall accuracy, average precision, average F1 score

establishes our anomaly-based technique as a valuable tool for fortifying computer systems and networks against emerging cyber threats.

In conclusion, this innovative strategy, incorporating a range of ensemble methods and robust evaluation metrics, lays the foundation for future advancements in adaptive, machine-learning-driven network intrusion detection. The success of our anomaly-based technique marks a significant contribution to the ongoing efforts to secure digital infrastructures. As we move forward, the adaptability and accuracy demonstrated in this research hold promise for shaping the future landscape of network security (Fig. 2).

Abbreviations and Acronyms

NIDS	Network Intrusion Detection System
CICIDS	2017 Dataset for network intrusion
DNN	Deep Neural Network
GAN	Generative Adversarial Network
FPR	False Positive Rate
DdoS	Distributed Denial of Service

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